MetaMed VR: Enhancing Medical Education through Immersive Virtual Reality

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Student's Declaration

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Shoaib Akhtar Signature: _____

_____

Verified by Plagiarism Cell Officer

Dated:

Certificate of Approval

The Department of Computer Science, National University of Computer and Emerging Sciences, accepts this thesis titled *MetaMed VR: Enhancing Medical Education through Immersive Virtual Reality*, submitted by Dawood Sarfraz (20P-0153), Waleed Akram (20P-0640), and Shoaib Akhtar (20P-0147), in its current form, and it is satisfying the dissertation requirements for the award of Bachelors Degree in Bachelors in Computer Science.

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Abstract

The rapid evolution of technology continues to redefine traditional educational paradigms. This report delves into the pioneering exploration of leveraging Immersive Virtual Reality (VR) to transform medical education. Focused on the project "Enhancing Medical Education through Immersive Virtual Reality (MetaMed)," our study aims to revolutionize the pedagogical landscape of medical training.

In the initial phase of our project, we embarked on a journey to harness cutting-edge VR technology. The objective was to create immersive learning environments that simulate real-world medical scenarios, allowing students to engage in interactive, hands-on learning experiences. Through the integration of advanced VR techniques, our project aspires to bridge the gap between theoretical knowledge and practical application, fostering a deeper understanding of medical concepts.

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Chapter 1

Introduction

- A vast and connected virtual universe where people can interact, socialize, work, learn and participate in various activities using technology is what is meant by the term "metaversion". This digital world creates a shared, immersive and often three-dimensional space by combining elements of Virtual Reality (VR), Augmented Reality (AR) and the Internet. The word "metaverse" comes from the prefix "meta-" (meaning beyond or beyond) and the word "universe" and refers to an area outside of our physical reality where people can interact, explore, and create their online presence.

- At the heart of the metaversion are immersive environments that users can explore and interact with through digital avatars. Users can customize these avatars to reflect their identity, tastes and physical characteristics. Metaverse offers a multi-sensory, interactive environment that goes beyond the possibilities of conventional 2D interactions on the Internet and simulates the experience of the real world in a digital environment.

- In addition to entertainment and social life, metaversion has important implications for work and education. Integrating virtual offices, classrooms and training environments into the metaverse opens up new possibilities for learning, skill development and remote collaboration.

1.1 History of the Metaverse

Also known as collective virtual shared space or virtual reality space, the idea of metaversion has a long and evolving history. Science fiction, technology and online communities are its ancestors. Here's a brief overview of the meta version's history:

1.1.1 1970 - Sci-Fi Origins:

The concept of metaversion can be traced back to works of science fiction. Notably, Neal Stephenson's 1992 novel "Snow Crash" introduced the idea of the Metaverse, a virtual reality space where users interacted as avatars. This novel popularized the term and served as a conceptual basis for later developments.

1.1.2 Early virtual worlds:

In the late 1970s and 1980s, online environments such as MUDs (Multi-User Dungeons) and MOOs (MUDs, Object-Oriented) allowed multiple users to interact in text-based virtual worlds. These were the forerunners of modern virtual worlds and online communities.

1.1.3 1990 - Emergence of 3D virtual worlds:

The 1990s saw the emergence of 3D virtual worlds such as Second Life, which allowed users to create avatars, socialize and build their virtual environments. These worlds laid the groundwork for more immersive metaverse experiences.

1.1.4 2000 - Online Games and Social Media:

Online interactions and communities have begun to shape massively multiplayer online games (MMOs) such as World of Warcraft and social media platforms such as Facebook and Twitter, bringing people closer to the idea of connected digital spaces.

1.1.5 2010 - VR and AR:

The development of virtual reality (VR) and augmented reality (AR) technologies has expanded the possibilities of metaversion. Platforms like the Oculus Rift, HTC Vive and the rise of AR via smartphones and AR glasses have begun to make immersive virtual spaces available.

1.1.6 Metaverse Investment:

In the mid-2010s, technology companies began investing heavily in metaversion-related projects. For example, Facebook acquired Oculus VR in 2014 and renamed it Meta in 2021, emphasizing its commitment to building the metaverse.

1.1.7 Vision of the Metaverse:

Companies like Meta (formerly Facebook), Epic Games, and other tech giants have expressed ambitious visions of the metaverse, envisioning a seamless, connected virtual universe where people can work, play, socialize, and engage in various activities.

1.1.8 Current developments:

As of my graduation date in September 2021, various companies are actively developing meta-related technologies such as virtual reality, augmented reality, blockchain, and decentralized platforms. NFTs (Non-Fungible Tokens) have also been used to represent digital assets within the metaverse.

1.2 The Future of the Metaverse

The future of metaversion holds the promise of widespread adoption and integration across various sectors with increasingly immersive virtual and augmented reality experiences, connected virtual worlds, blockchain-based asset ownership, and economic op-

portunities in digital realms. It will impact society by redefining social dynamics and cultural norms while raising concerns about privacy, security, regulation and environmental sustainability. Artificial intelligence will improve user experience and the metaverse will serve as a platform for education, healthcare, entertainment, gaming, work and collaboration, fundamentally changing the way we interact with the digital realm and each other.

1.3 How Metaverse Would Excel

1.3.1 What would affect the success of the metaverse

The success of the metaverse will be influenced by several key factors: **Technology Advances:** The success of the metaverse depends on the continuous development and advancement of technologies such as virtual reality (VR), augmented reality (AR), blockchain, artificial intelligence (AI), and high-speed internet connectivity. Improved hardware and software are essential to create immersive and seamless virtual experiences. **User Adoption:** The success of the metaverse depends on a critical mass of users and active participants. The more people get involved and contribute to the metaversion, the more valuable and alive it becomes. User-friendly interfaces and accessibility will play a key role in gaining a diverse user base. **Content and Creativity:** A wide variety of engaging content, experiences, and activities within the Metaverse is essential. Content creators, developers and users need to be able to easily create and share experiences. This includes virtual worlds, games, social spaces, art, educational content and more. **Interoperability and standards:** Interoperability between different virtual worlds and platforms is key to a seamless metaverse. Developing open standards and protocols for the metaverse can help create a more connected and accessible environment. **Security and Privacy:** Users must trust metaverse with their personal data and transactions. Robust security measures, data protection and privacy controls are vital to prevent abuse, fraud and unauthorized access. **Economic Viability:** A sustainable economy within the metaversion will be necessary to attract users and content creators, including ownership of virtual real estate, digital cur-

rencies, and employment opportunities. Blockchain technology and non-fungible tokens (NFTs) can play a role in creating digital property rights and assets. **Regulation and governance:** Governments and regulators will need to create clear guidelines and legal frameworks for meta

Chapter 2

Literature overview

The integration of metaversion, a virtual, interconnected digital realm, into the healthcare field represents a promising frontier in medical technology and patient care. As technology continues to advance, metaverse offers innovative solutions to long-standing challenges in the healthcare industry, from medical training and telemedicine to mental health therapy and patient empowerment. This literature review aims to provide a comprehensive examination of current research, technology and developments in the use of metaversion in healthcare. By examining the historical perspective, technological foundations, practical applications, ethical considerations, and future prospects of metaversion in healthcare, this review seeks to illuminate the current state of knowledge and pave the way for further advances in this rapidly evolving field.

2.1 Everything you need to know about the metaverse: A complete exploration of the technological singularity, virtual ecosystem and research agenda

2.1.1 Posts:

The article offers a comprehensive framework for understanding the evolution of metaversion, exploring state-of-the-art technologies and metaversion ecosystems. It identifies eight core technologies, including augmented reality, artificial intelligence, and blockchain, that are driving the transition from the current internet to the meta versions. This paper discusses six user-centric factors such as avatar, content creation, and virtual economy that are essential to creating a self-sustaining and shared empire within the metaverse. It proposes a specific research agenda for the development of the metaverse and provides a blueprint for future studies and innovations in the field.

2.1.2 Limitations:

The article does not provide an in-depth analysis of each enabling technology and user-centric factor, but rather offers a broad overview of their importance to the evolution of the metaverse. The research agenda proposed in this document may not cover all possible research directions and may require further refinement based on evolving technologies and user needs. The article does not extensively address the potential challenges and risks associated with metaversion, such as ethical considerations, inequality, and the impact on physical reality. The article does not include empirical studies or case studies to support its claims and frameworks, which rely primarily on a survey-based approach. [2]

2.2 Health Care in the Metaverse: A Survey of Current Applications of the Metaverse v Health care

[1]

2.2.1 Posts:

The article presents a comprehensive overview of metaverse applications in the healthcare industry, covering seven areas: telemedicine, clinical care, education, mental health, physical fitness, veterinary medicine, and pharmaceuticals. It offers an exploration of the latest developments in metaverse technology and their potential to provide alternative health care systems. The article discusses the technical challenges and available solutions in each domain and provides insight into the challenges and opportunities of implementing metaverse applications in healthcare. It highlights the need for a self-sustaining, sustainable and forward-looking solution for medical healthcare systems using metaverse technology. The paper identifies the challenges that need to be addressed before the full adoption of metaversion for the healthcare industry, highlighting the importance of connectivity, privacy, security, integration, interoperability, user experience and technical issues related to VR and AR technologies.

2.2.2 Limitations:

Connection and bandwidth limitations may affect real-time interaction and communication in metaverse healthcare applications. Privacy and security concerns arise due to the sensitive nature of health data in the metaverse. Ensuring data protection and secure communication channels is essential. Integration and interoperability issues may arise when meta-version applications are integrated with existing healthcare systems and infrastructure. Ensuring seamless data exchange and compatibility is important. User experience and interface design must be optimized to provide a user-friendly and intuitive environment in metaverse healthcare applications. Technical issues related to Virtual Reality

(VR) and Augmented Reality (AR) technologies, such as motion sickness, latency, and device compatibility, need to be resolved for a smooth and immersive experience for the user.

2.3 Medical Metaverse: Technology, Applications, Challenges and Future

[3]

2.3.1 Contributions:

An overview of metaversion technologies and applications in the medical field. Exploring the potential of metaversion in healthcare through various application scenarios. Analysis of current challenges in implementing medical metaversion and proposing possible solutions. Emphasizing the importance of continuous improvement in medical experience and quality in addressing shortages of medical resources and high costs of medical care. Emphasizing the immersive experience and efficiency that the metaverse can provide to medical staff in terms of diagnosis, education and treatment. Recognizing the strong interaction between medical staff and patients in the digital space of metaversion.

2.3.2 Future work:

Further exploration of the ethical and privacy implications of using metaversion in healthcare, consideration of potential risks and safeguards required for patient data and confidentiality. Conducting in-depth technical feasibility studies and implementation challenges of integrating various technologies into the medical meta-version, addressing issues such as interoperability, data security, and system integration. Conducting empirical research and case studies to evaluate the effectiveness and efficiency of using metaversion in healthcare settings, providing evidence-based insights into its impact on patient

outcomes, healthcare delivery and cost-effectiveness. Exploring potential disparities in access to the medical metaverse, particularly for underserved populations or regions with limited technological infrastructure, and devising strategies to bridge the digital divide. Exploring the potential for collaboration and partnership between health care providers, technology developers, and regulators to develop standards, guidelines, and best practices for the safe and effective use of medical metaversion.

2.4 The Metaverse for Healthcare: A Survey of Potential Applications, Challenges, and Future Directions.

[4]

2.4.1 Contributions:

We provide a comprehensive review of Metaverse for Healthcare, highlighting the state of the art and potential applications. Identifying technologies enabling Metaverse adoption in healthcare such as artificial intelligence, virtual reality

Chapter 3

Project Vision

This chapter should contain the following information.

3.1 Problem Statement

Medical students often face challenges in gaining hands-on medical experience due to limited availability of actual medical cases, ethical concerns, and high costs associated with traditional medical training methods. This project aims to address these challenges by leveraging the metaverse to create a virtual medical training platform that provides realistic medical scenarios and opportunities to practice in a risk-free environment.

3.2 Business Opportunity

The business opportunity is to offer an innovative, cost-effective and affordable medical training solution to medical students and institutions. A metaverse-based medical training platform can potentially reduce the costs associated with traditional medical training and expand access to high-quality training experiences, thereby improving the skills and readiness of future physicians.

3.3 Goals

The main goals of this project are: Develop a metaverse-based medical training platform that accurately simulates medical procedures. To provide medical students with a risk-free environment to practice medical skills. To enhance the learning experience and improve medical competence among medical students. Reduce costs and ethical concerns associated with traditional medical training methods.

3.4 Project Scope

The project will involve the development of a metaverse-based medical training environment that will cover a range of medical procedures across different medical specialties. It will include creating realistic medical simulations, integrating haptic feedback devices for tactile training and providing medical students with access to hands-on practice. The project will primarily focus on the technical and educational aspects of metaverse integration.

3.5 Limitations

Constraints include budgetary limitations for technology development and software integration, technology limitations related to metaverse platforms, and the need to adhere to medical training and ethical standards. The project will also need to operate within a set time frame and must comply with relevant regulations.

3.6 Description of Stakeholders

Key stakeholders in this project include medical students, physician educators, healthcare institutions, and potentially patients who may benefit from better trained medical professionals.

3.6.1 Stakeholder Summary

Medical students are the primary beneficiaries and users of the metaverse-based medical training platform. They can gain valuable hands-on experience that is often limited in traditional training. Medical educators are stakeholders involved in developing educational content and ensuring compliance with medical curriculum standards. Healthcare institutions can support or adopt the metaverse training platform to improve the skills and readiness of medical students and residents.

3.6.2 Key High Level Objectives and Stakeholder Issues

Medical students seek practical medical training to supplement their theoretical knowledge and gain expertise in various medical procedures. The goal of medical educators is to provide an effective and safe medical training environment that is consistent with the requirements of the medical curriculum. Healthcare facilities are interested in producing well-trained medical professionals who are able to provide quality care.

Chapter 4

Software Requirements Specification

This chapter will contain the functional and non-functional requirements of the project.

4.1 List of functions

List of key features for the "Improving Medical Education through Immersive Virtual Reality (MetaMed)" project:

Realistic medical simulations: Create highly realistic medical scenarios and procedures that faithfully mimic real medical experiences, including various medical specialties.

Haptic feedback integration: Integrate tactile feedback devices to provide tactile sensations, increasing the realism of medical training and allowing students to develop their tactile skills.

Immersive 3D Environments: Create immersive, 3D virtual environments that accurately represent medical settings, including operating rooms, medical instruments, and anatomical structures.

Interactivity: Enables hands-on interaction with virtual medical tools, instruments, and equipment, allowing students to perform medical tasks and procedures.

Remote Access: Enable medical students to access a metaverse-based medical training platform from remote locations, facilitating flexible learning.

4.2 Test plan (test level, test techniques)

Test plan for our project will be as:

- Movement of Student in VR
- Interaction with Objects
- Treatement to Patient

4.3 Functional Requirements

4.4 Use Case Diagram

Here is the Use Case Diagram of the Project.

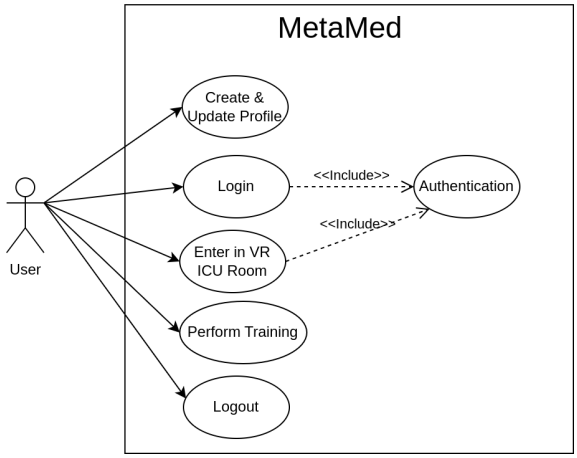


Figure 4.1: Use Case Diagram

4.5 Software Development Plan

Software Development Plan for our project will be as:

- Project Overview
- Requirements Gathering and Analysis
- System Design
- Development
- Testing
- Documentation
- Management

4.6 Sequence Daigram/System Diagram

Here is the Sequence Daigram/System Diagram of the Project.

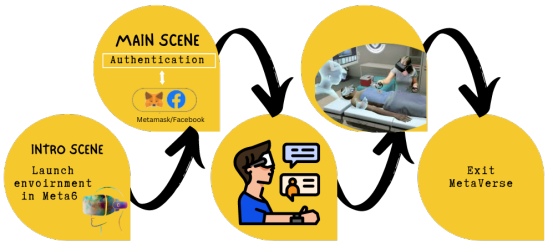


Figure 4.2: System Diagram

4.7 Activity Daigram

The activity diagram aims to illustrate the entire process of project working and demonstration.

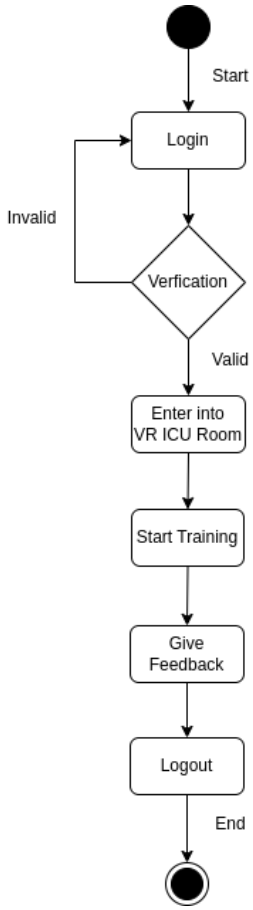


Figure 4.3: Activity Diagram

4.8 Tools used

- Blender
- Unity
- Microsoft Visual
- C#/C sharpe
- VRTK (Virtual Reality ToolKit)

Chapter 5

Iteration Plan

5.1 Planning Strategy

Planning Strategy for our project will be as:

5.1.1 FYP 1 Mid:

- Literature review related Metaverse, and Healthcare
- Designed Use-case, System Diagram

5.1.2 FYP 1 Final:

- Environment setup in Unity3D and Blender
- Designed basic environment of Hospital like Rooms and Cadavers
- Delivered Demo of our environment

5.1.3 FYP 2 Mid:

- User Movements

- Patient Information
- Medicines Information
- Basic Tools like syrings

5.1.4 FYP 2 Final:

- Debugging
- Oculus Integration
- Oculus Testing

5.2 UI/UX

UI/UX of Project.

Figure 5.1: UI/UX Diagram

Chapter 6

Iteration 1 (FYP-1 Mid)

- FYP 1 Mid:
  - Literature review related Metaverse, and Healthcare
  - Designed Use-case, System Diagram

6.1 Literature review related Metaverse, and Healthcare

The aof metaversion, a virtual, interconnected digital realm, into the healthcare field represents a promising frontier in medical technology and patient care. As technology continues to advance, metaverse offers innovative solutions to long-standing challenges in the healthcare industry, from medical training and telemedicine to mental health therapy and patient empowerment. This literature review aims to provide a comprehensive examination of current research, technology and developments in the use of metaversion in healthcare. By examining the historical perspective, technological foundations, practical applications, ethical considerations, and future prospects of metaversion in healthcare, this review seeks to illuminate the current state of knowledge and pave the way for further advances in this rapidly evolving field.

6.2 Designed Use-case, System Diagram

- Use Case
- Sequence Diagram/ System Diagram
- Activity Diagram

6.3 Use Case Diagram

A use case diagram represents the main use case activities and the interaction of actors and the system that is under development process. It helps to identify all main processes of the system which are then visualized in ovals, known as a use case. A use case diagram is drawn from a scenario that explains the working of the system.

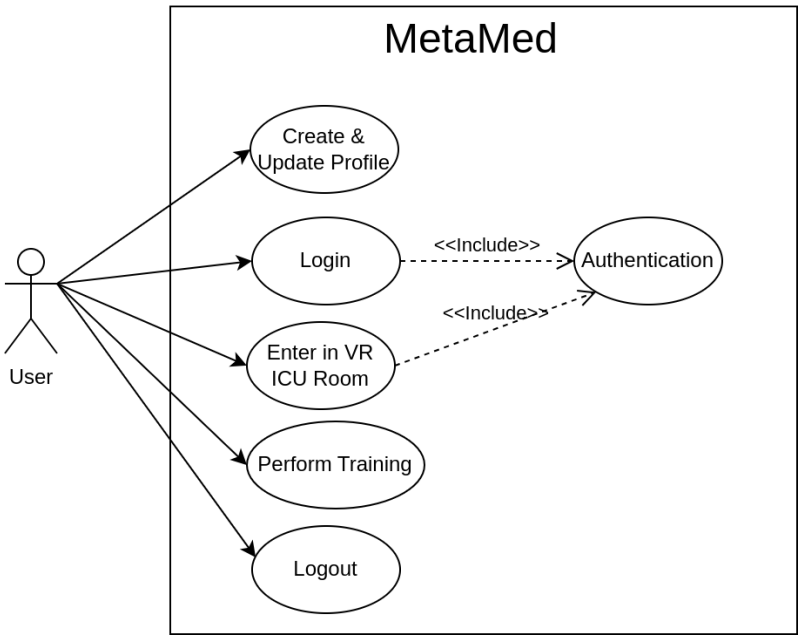


Figure 6.1: Use Case Diagram

6.4 Sequence Daigram/System Diagram

System Diagrams are models used to visually express the dynamic forces acting upon the components of a process and the interactions between those forces. System Diagrams are more than process flow charts.

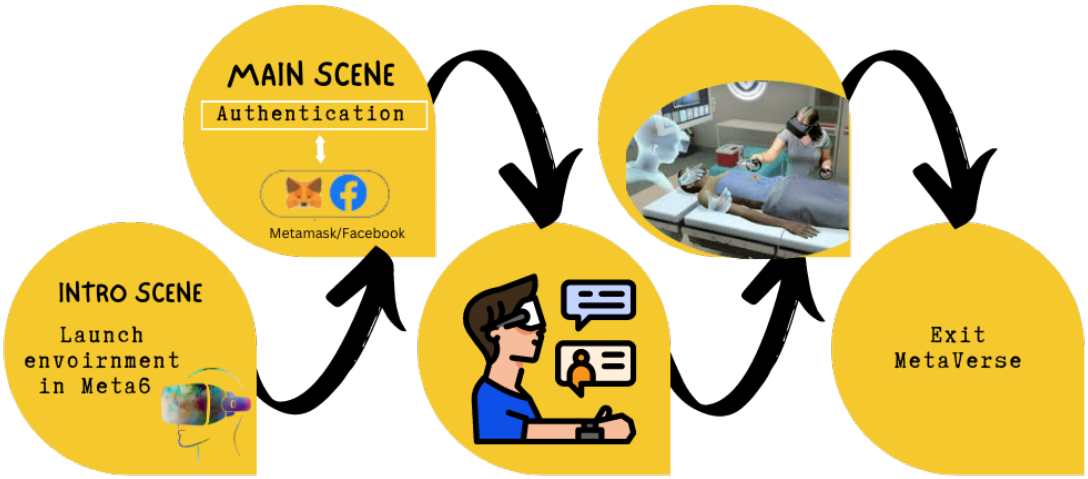


Figure 6.2: System Diagram

6.5 Activity Diagram

Activity diagram is a visual representation of a system's behavior, showing how activities and actions are executed in a particular process or use case. Activity diagrams are used to model a variety of systems and processes, including business workflows, software systems, and organizational processes.

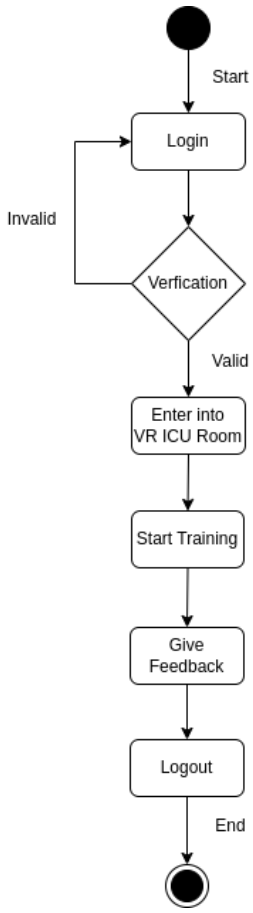


Figure 6.3: Activity Diagram

Chapter 7

Iteration Plan 2 (FYP 1 Final)

The second iteration plan for our project is explained in this chapter. This chapter will provide guidance on the project's modules and their development. Students are expected to talk about the project's second stage of method of implementation in this chapter:

7.1 FYP 1 Final:

- FYP 1 Final Presentation:
  - Environment setup in Unity3D and Blender
  - Designed basic environment of Hospital

7.2 Designed basic environment of Hospital

- Reception Area
- Waiting Rooms
- Patient Rooms
- Training Facilities

7.2.1 Reception Area

The reception area serves as the main entrance point to the virtual hospital. It features a welcoming reception desk where virtual receptionists or automated avatars greet patients and visitors. Patients can check-in here, provide necessary information, and receive directions to their appointments or designated areas within the hospital.

Figure 7.1: Reception Area

7.2.2 Waiting Area:

Waiting rooms are designed to mimic those found in real hospitals, providing comfortable seating arrangements for patients and their companions while they wait for appointments or procedures. These areas may include digital screens displaying health information, magazines, and interactive features to help patients pass the time.

Figure 7.2: Waiting Area

7.2.3 Patient Rooms:

Patient rooms are designed to provide a comfortable and healing environment for patients during their hospital stay. These virtual rooms feature adjustable beds, bedside tables, medical monitors, and communication devices for patients to interact with healthcare staff. Patients can use virtual interfaces to request assistance, access entertainment options, or adjust room settings according to their preferences. Training facilities within the

Figure 7.3: Patient Room

VR hospital environment offer educational resources and simulation modules for medical students, residents, and healthcare professionals. Overall, our VR hospital environment aims to replicate the real-world hospital experience while leveraging the immersive capabilities of virtual reality to enhance patient care, medical training, and healthcare delivery. By providing a realistic and interactive environment, we strive to improve patient outcomes, optimize clinical workflows, and advance medical education and research.

Chapter 8

Iteration 3 (FYP-2 Mid)

The first iteration is expected to be completed by the midterm of the FYP-2. This chapter will have some of the artifacts based on system design. The requirements analysis section is same for all the systems while the design may vary. There may have two types of designs the structural design or . First section is for the structural design.

8.1 FYP-2 Mid

Objectives for FYP-2 Mid

- Laboratory
- Add basic tools related to Medical
- User movements
- Patient Information
- Medicines Information

8.1.1 Laboratory

We added Laboratory where student can find basic tools and medicines.

Figure 8.1: Laboratory Room

8.1.2 Patient Information

We have stored Patient Information in XML file. It will help us to keep record of Patient.


```xml
  <Patient mID="1">
    <mName>Jawad Ali</mName>
    <mAge>32</mAge>
    <mWeight>70</mWeight>
    <mType>Adult</mType>
    <mSex>Male</mSex>
      <mAllergies />
  </Patient>
  <Patient mID="2">
    <mName>Salman</mName>
    <mAge>19</mAge>
    <mWeight>65</mWeight>
    <mType>Child</mType>
    <mSex>Male</mSex>
       <mAllergies />
  </Patient>
  <Patient mID="3">
    <mName>Sultan</mName>
    <mAge>72</mAge>
    <mWeight>70</mWeight>
    <mType>Elder</mType>
    <mSex>male</mSex>
       <mAllergies />
  </Patient>
</mPatients>
```


Figure 8.2: Patient Information

8.2 Medicines Information

We have stored Medicine Information in XML file. It will help us to keep record which medicine would be used given Symptoms.

```xml
<mMedicines>
  <Medicine mID="0">
    <mName>primperan#gray</mName>
    <mQuantity>6</mQuantity>
    <mUnit>ml</mUnit>
    <mPackage>flask</mPackage>
    <mPointsOfAttention>When feel sick, have a headache, or throw up.</mPointsOfAttention>
  </Medicine>
  <Medicine mID="1">
    <mName>lltlcan#gray</mName>
    <mQuantity>6</mQuantity>
    <mUnit>ml</mUnit>
    <mPackage>flask</mPackage>
    <mPointsOfAttention>"When feel sick, have a headache, or throw up."</mPointsOfAttention>
  </Medicine>
  <Medicine mID="2">
    <mName>dipidolor#gray</mName>
    <mQuantity>6</mQuantity>
    <mUnit>ml</mUnit>
    <mPackage>flask</mPackage>
    <mPointsOfAttention>"Dipidolor is administered before, during, and/or after surgery to reduce pain."
    </mPointsOfAttention>
  </Medicine>
  <Medicine mID="3">
    <mName>paracetamol</mName>
    <mQuantity>16</mQuantity>
    <mUnit>units</mUnit>
    <mPackage>box</mPackage>
    <mPointsOfAttention>"Painkiller and helps with fever."</mPointsOfAttention>
  </Medicine>
  <Medicine mID="4">
    <mName>lormetazepam</mName>
    <mQuantity>50</mQuantity>
    <mUnit>units</mUnit>
    <mPackage>box</mPackage>
    <mPointsOfAttention>"For treating insomnia."</mPointsOfAttention>
  </Medicine>
```


Figure 8.3: Medicine Information

8.3 SVN or GitHub

We have uploaded on GitHub. Here is the GitHub link:

https://github.com/Daudsarfraz/MetaMed

Chapter 9

Iteration 4

The first iteration is expected to be completed by the final of the FYP-2. This chapter will have some of the artifacts based on system design. The requirements analysis section is same for all the systems while the design may vary. There may have two types of designs the structural design or . First section is for the structural design.

structural design

9.1 Domain Model/ Class Diagram

9.2 Component Diagram

Behavior Design

9.3 Flow Diagram

9.4 Data Flow Diagram (DFD)

9.5 Data Dictionary

9.6 Activity Diagram

9.7 Network Automata/ Graphs or State Machine

9.8 Call Graph or Sequence Diagram

9.9 Interaction Overview Diagram

For all above designs

9.10 Schema Design/ ER Diagram

9.11 Data Structure Design

Any information

9.12 Algorithm Design

Any information

9.13 Development Phase

Comments, Naming Conventions, Static Analysis of Code, etc.,

9.13.1 Unit Test

9.13.2 Suites or Test Cases

9.14 Maintainable Phase

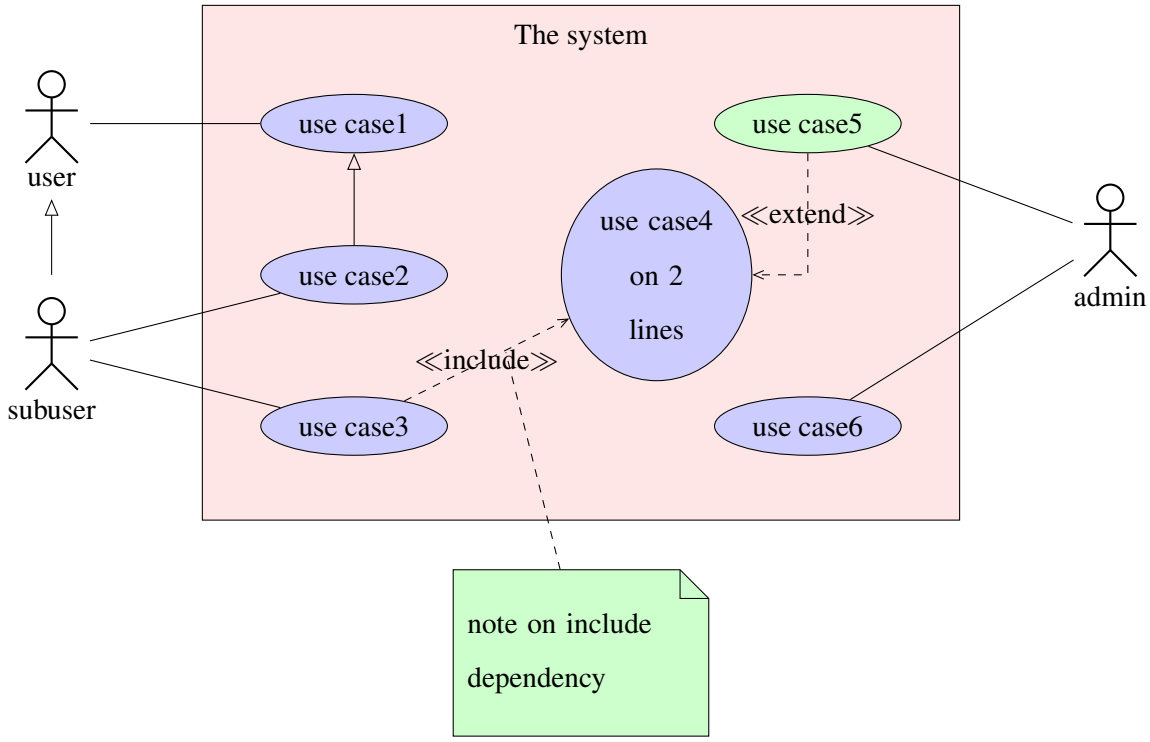
9.14.1 CI/ CD

9.14.2 Deployment Diagram

9.14.3 System-Level Test Suites, Test Cases

9.14.4 SVN or GitHub (Optional)

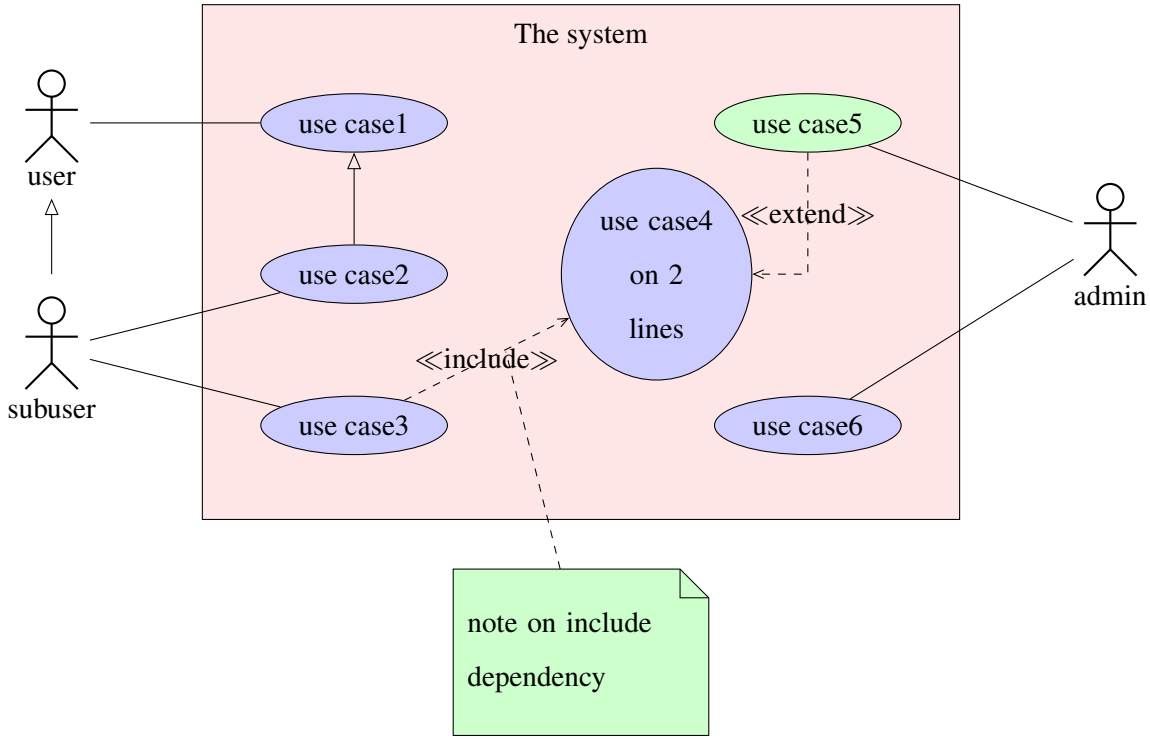
9.14.5 Configuration/ Setup and Tool Manual (Optional)



Chapter 10

Implementation Details

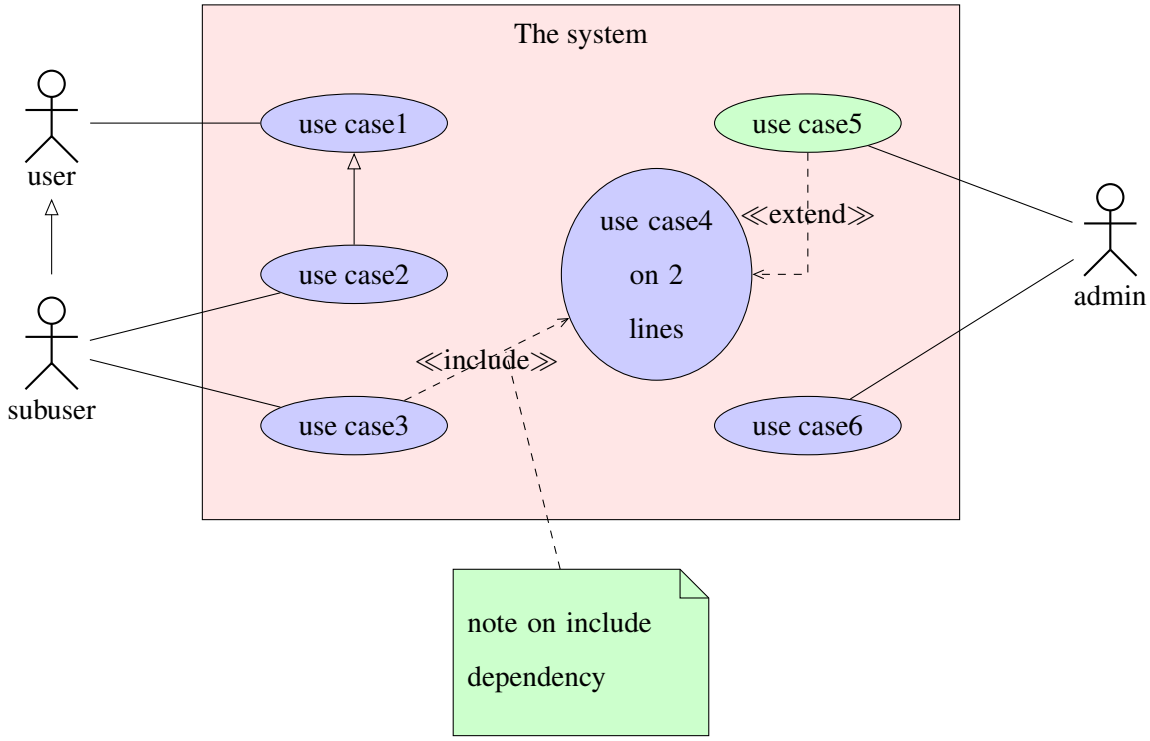
not the programming code but the algorithmic and procedural details especially related to the hidden/ backend algorithms that are not covered in the design



Chapter 11

User Manual

This chapter will have the user manual.



Chapter 12

Conclusions and Future Work

conclusions here

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